

Source: https://en.wikipedia.org/wiki/Operating_system

25CS208 OPERATING SYSTEMS

Hours Per Week :	L	T	P	SL	C
	2	2	0	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization & Digital logic and its design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility in acting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1

LINUX FILE SYSTEM

Introduction to LINUX File System: The LINUX file System, File System Hierarchy, File system Commands, File Attributes, File Permissions.

Basic commands: cal, date, echo, bc, script, passwd, who, uname, tty, stty, pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, more, file, wc, od, chmod, vi. (T1, Pg. No: 45 - 78)

Filters: cmp, comm, diff , head, tail, find, cut, paste, sort, uniq. (T1, Pg. No: 95 - 96)

Regular Expressions: grep, egrep, fgrep, Sed- line addressing, context addressing, text editing, substitution. (R4, Pg. No: 354 – 356, 409)

Shell Script: Creating shell script, Shell Input, Shell Output, Basic operators, Conditional and Iterative statements. (R4, Pg. No: 547 – 567)

UNIT-2

PROCESS SCHEDULING

Introduction to Operating System: What Operating System do; Operating System Structure; (T2, Pg. No: 03) Operating System Operations: Process Management, Memory Management, Storage Management. (T2, Pg. No: 18 – 29)

System Security: Security Problem, Program Threats, System and Network Threats, User Authentication. (T2, Pg. No: 621 – 650)

Process concept-overview, Process Scheduling, Operations on Process; Inter Process Communication; (T2, Pg. No: 101 – 123)

Process (CPU) Scheduling-Scheduling Criteria, Scheduling Algorithms; Multiple-Processor scheduling; (T2, Pg. No: 183 – 205)

PRACTICES:

- Use the cat command to create a fi le containing the following data. Call it mytable.txt use tabs to separate the fields.
- 1425 ravi 15.65
- 4320 ramu 26.27
- 6830 sita 36.15
- 1450 raju 21.86

a. Use the cat command to display the fi le, mytable.txt.

b. Use the vi command to correct any errors in the fi le, mytable.txt.

c. Use the sort command to sort the fi le mytable.txt according to the first field.

d. Call the sorted fi lemytable.txt (same name)

e. Print the fi lemytable.txt.

f. Use the cut &paste; commands to swap fields 2and 3mytable.Call itmytable.txt (same name)

g. Print the new fi le, mytable.txt.

- Write a shell script that takes a command–line argument and reports on whether it is directory, a file, or something else.
- Write a shell script that accepts one or more file name as arguments and converts all of them to uppercase, provided they exist in the current directory.
- Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- Write a shell script that computes the total and average marks of a student according to the following;
- If average marks ≥ 69 then result is -Distinction.
- If average marks ≥ 59 and ≤ 70 then result is – First Class.
- If average marks ≥ 49 and ≤ 60 then result is – Second Class
- If average marks ≤50 then result is -Pass.
- Note that any subject marks ≤ 40 then result is -Fail.
- Accept student name and six subject marks through the keyboard.
- Write an interactive fi le-handling shell program. Let it offer the user the choice of copying, removing, renaming, or linking files. Once the user has made a choice, have the program ask the user for the necessary information, such as the fi le name, new name and so on.
- Write a shell script, which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second fi le should be deleted.
- Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
- Write a shell script to biggest among three numbers.
- Write a shell script to find whether the entered number is prime or not.
- Write a shell script to find the whether the entered number is Armstrong or not.
- Write a shell script to find the factorial of the given number.
- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.

MODULE-2

UNIT-1

PROCESS SYNCHRONIZATION & DEADLOCKS

Process Synchronization: The critical-section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization: Producer-Consumer Problem, Readers-Writers Problem, Dining Philosophers Problem, Monitors. (T2, Pg. No: 225 – 252)

DEADLOCKS

Deadlocks: Deadlock characterization; Methods of handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery. (T2, Pg. No: 285 – 305)

UNIT-2

MEMORY MANAGEMENT

Memory Management: Basic concept to memory management, Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation. (T2, Pg. No: 315 – 345)

Virtual Memory Management: Demand Paging, Page Replacement: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU), Least Recently used (LRU), Allocation of Frames. (T2, Pg. No: 357 – 385)

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

UNIT-3

SECONDARY STORAGE STRUCTURE

Secondary Storage Structure: Over view of mass-storage structure, Directory & Disk Structure, RAID Structure, disk scheduling; (T2, Pg. No: 505 – 532)

File System Interface - File concept, File-system structure, Allocation Methods, Access Methods, File-System Mounting, File Sharing, Protection, Free Space Management. (T2, Pg. No: 461 – 482)

PRACTICES:

- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Dead lock avoidance.
- Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.

a) What is a lower bound on the number of page faults?

b) What is an upper bound on the number of page faults?

- Consider the following page-replacement algorithms. Rank these algorithms on a five-point scale from "bad" to "perfect" according to their page-fault rate. Separate those algorithms that suffer from Belady's anomaly from those that do not.

a) LRU replacement. b) FIFO replacement. c) Optimal replacement. d) Second-chance replacement.

- Develop the program to find the total no. of page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames for the gives reference string?
- Page reference string: 1, 2, 3, 4,2, 1, 5, 6, 2, 1, 2, 3,7, 6, 3, 2, 1, 2, 3, 6.
- Remember that all frames are initially empty, so your first unique pages will cost one fault each.
- LRU replacement.
- FIFO replacement.
- Optimal replacement.
- What is the minimum number of page faults for an optimal page replacement strategy for The reference string 1, 2, 3,4, 5, 3, 4, 1, 6,7, 8, 7, 8, 9, 7, 8,9, 5, 4, 5,4, 2. with four page frames?
- Consider a demand-paged computer system where the degree of multiprogramming is currently fixed at four. The system was recently measured to determine utilization of the CPU and the paging disk. Three alternative results are shown below. For each case, what is happening?

a) Can the degree of multiprogramming be increased to increase the CPU utilization? Is the paging helping?

b) CPU utilization 13 percent; disk utilization 97 percent.

c) CPU utilization 87percent; disk utilization 3 percent.

d) CPU utilization 13 percent; disk utilization 3 percent.

- Implementation of Disk scheduling algorithm–FCFS.
- Implementation of Disk scheduling algorithm–SSTF and SCAN.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply regular expressions and LNIX text processing tools to search, filter, manipulate text data and develop the shell programming.	Apply	M – 1, U - 1	PO1
2	Apply the concepts of process scheduling algorithms and process synchronization techniques to derive the efficiency of resource utilization.	Apply	M – 1, U - 2	PO1, PO2, PO3
3	Analyze process synchronization issues and evaluate deadlock conditions to identify and recommend appropriate solutions for safe process execution.	Analyze	M – 2, U - 1	PO1, PO2, PO4

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
4	Apply memory and virtual memory management techniques such as paging, segmentation, and page replacement algorithms to manage and optimize system memory usage.	Apply	M – 2, U – 2	PO3, PO5, PO7, PO9
5	Analyze secondary storage structures and file system designs to evaluate the performance and efficiency of disk scheduling algorithms, directory organization, and space management techniques.	Analyze	M – 2, U – 3	PO2, PO4, PO7, PO11
6	Evaluate the performance of different memory allocation and replacement algorithms under varying workloads and system conditions.	Evaluate	M – 2, U - 2	PO2, PO4, PO6, PO7, PO11

MAPPING OF SDGS AND IKS

Unit	Topic	Sustainable Development Goals (SDGs)	Indian Knowledge System (IKS)
M – I Unit – I	Linux commands, Filters, Regular Expressions, Shell Programming.	SDG 4 – Quality Education SDG 9 – Industry, Innovation & Infrastructure	Integrate IKS with Modern Education, Promote Awareness and Understanding
M – I Unit – II	Basics of Operating Systems, Operating system security, Process, Process/ CPU scheduling.	SDG 4 - Quality Education, SDG 9 – Industry, Innovation & Infrastructure, SDG 12 – Responsible Consumption & Production	Encourage Research and Innovation, Build Capacity and Expertise
M – II Unit – I	Process Synchronization, Critical section Problem, Semaphores, Monitors, Deadlocks, Handling deadlocks, Deadlock detection and recovery.	SDG 9 - Industry, Innovation & Infrastructure, SDG 12 - Responsible Consumption & Production	Encourage Research and Innovation, Preserve and Promote Indigenous Knowledge
M – II Unit – II	Memory management, Paging, Segmentation, Virtual Memory management, Page replacement Techniques,	SDG 9 – Industry, Innovation & Infrastructure, SDG 8 – Decent Work and Economic Growth	Promote Awareness and Understanding, Encourage Translation and Documentation
M – II Unit – III	Mass storage, Disk structure, Disk Scheduling, File, File sharing, File mounting, File allocation methods, Directory implementation , Free space management	SDG 9 – Industry, Innovation & Infrastructure, SDG 8 – Decent Work and Economic Growth, SDG 11 – Sustainable Cities & Communities	Build Capacity and Expertise, Foster Global Recognition and Collaboration

TEXT BOOKS:

1. Sumitabha Das, Unix concepts and applications□, TMH Publications, 4th Edition, July 2017. 2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Concepts”, 9th Edition, John Wiley & SonsInc, 2013.

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REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, “Advanced Programming in the Unix Environment”,3rd Edition, Addison-Wesley, 2013. 2. William Stallings, “Operating Systems-Internals and Design principles” PHI, 7th Edition, 2012. 3. Gary J. Nutt. Addison-Wesley, “Operating Systems: A Modern Perspective”, 2nd Edition, 2001. 4. B.A. Forouzan & R.F.Giberg, - Unix and shell Programming□, Thomson, 1st Edition, New Delhi,2003.

2. William Stallings, “Operating Systems-Internals and Design principles” PHI, 7th Edition, 2012.

3. Gary J. Nutt. Addison-Wesley, “Operating Systems: A Modern Perspective”, 2nd Edition, 2001.

4. B.A. Forouzan & R.F.Giberg, - Unix and shell Programming□, Thomson, 1st Edition, New Delhi,2003.